

Table 1: notation used throughout the paper.

Symbol	Description
A	sender agent
R	receiver agent
S	set of all possible messages used for communication by both agents
O	set of mammal classes
Q	set of mammal images available to the sender
W	set of mammal descriptions available to the receiver
g	ground-truth map between the images and the descriptions
m	binary message sent by the sender
n	binary message sent by the receiver
T	maximal number of time steps in a conversation
t	time step in conversation between sender and receiver
h	hidden state vector of the sender
k	hidden state of the receiver at time step t
B	baseline feedforward network of the sender
L	per-instance reinforcement learning loss
H	entropy regularization coefficient for the binary message distributions